

Projections of Heat-Stress Conditions Across Bangladesh Using Explainable Machine Learning and Climate Trend Integration

Abstract

Rising heat and humidity increasingly threaten health, labour, and energy systems across Bangladesh, yet operational planning needs district-resolved, uncertainty-aware projections of the specific conditions that drive human heat stress. We present a deterministic, reproducible machine-learning framework that projects seven temperature-correlated heat-stress conditions, heat-index comfort, wet-bulb temperature, indoor and outdoor wet-bulb globe temperature, cooling degree-days, ultraviolet exposure, and dew-point moisture stress for all 64 districts of Bangladesh. Four gradient-boosting ensembles are trained on the 2014–2024 monthly record using future-known seasonal features; the observed 1980–2024 Theil–Sen trend is superimposed so that projections evolve realistically, and all thermal indices are recomputed from the projected primary variables to avoid target leakage. Validated against withheld 2025 observations, the projected indices attain median R^2 of 0.95–0.98, and their operational risk classes match observations to within one class in 100% of district-months. A Mann-Kendall analysis shows the maximum temperature rising in every district (median $+0.56^\circ\text{C}$ per decade) while minimum temperature remains flat, indicating a widening diurnal range; the coastal and southwestern zones carry the greatest combined burden, and ultraviolet and oppressive humidity hazards are effectively nationwide. District choropleths, zonal envelopes, and calibrated uncertainty bands provide a directly actionable heat-stress outlook while transparently bounding the models’ skill relative to climatology. The complete 64-district, seven-condition projection is openly explorable through an accompanying interactive web application.

Keywords: Temperature-Correlated Conditions, public health, Bangladesh, seasonal projection, Applied Machine Learning